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Reducing vortex losses in superconducting microwave resonators with microsphere patterned antidot arrays

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We experimentally investigate the vortex induced energy losses in niobium coplanar waveguide resonators with and without quasihexagonal arrays of nanoholes (antidots), where large-area antidot patterns have been fabricated using self-assembling microsphere lithography. We perform transmission spectroscopy experiments around 6.25 GHz in magnetic field cooling and zero field cooling procedures with perpendicular magnetic fields up to $B = 27$ mT at a temperature $T = 4.2$ K. We find that the introduction of antidot arrays into resonators reduces vortex induced losses by more than one order of magnitude. © *2012 American Institute of Physics*. [doi:10.1063/1.3673869]

The importance of superconducting microwave circuitry devices has continuously grown during the last years. In particular, coplanar microwave resonators have become very popular and are used in various fields such as circuit quantum electrodynamics,[1–3] quantum information processing,[4] and kinetic inductance particle detection.[5] As low energy losses are an essential requirement to these resonators, there are currently many efforts to identify and minimize the individual dissipation mechanisms.[6–8] Most recently, it has been proposed to couple superconducting devices to trapped magnetic molecules, single electrons, or ultracold atom clouds,[9–13] which requires superconducting low-loss circuitry to be operated in magnetic environments.[14,15] However, operating type-II superconducting resonators in magnetic fields leads to considerable energy dissipation due to Abrikosov vortex motion.[16] Recently, some approaches have been made to reduce the vortex associated losses under special experimental conditions. Amongst others,[17,18] patterning the resonators with antidots, well-known and highly controllable pinning sites for Abrikosov vortices,[19,20] has shown promising results.[21] It was demonstrated that a few antidots with diameters on the micron scale, strategically positioned at the resonator edges, are able to significantly reduce vortex associated losses in zero field cooling (zfc) experiments with magnetic fields of ~10 G. In addition, we found that the losses decreased with increasing number of antidots. Thus, in order to expand the magnetic field range for the operation of the resonators, it is necessary to increase the antidot density and, at the same time, reduce the antidot diameter. In the past, there have been various approaches based on self-assembling techniques,[22–24] as they allow to cover the whole substrate area with a large amount of tiny antidots. Yet, for high-frequency applications, these techniques cannot be easily applied, as some of them depend on the substrate material[23,25] itself and most of them change the substrate properties during the fabrication process significantly.

In this letter, we adopt a different fabrication technique and demonstrate it to be feasible for patterning large area antidot lattices into superconducting microwave resonators. We use self-assembled polystyrene colloids as a quasi-hexagonal array of microlenses which can be used for the patterning of antidots by optical lithography.[26] Such patterned samples show vortex associated microwave losses in perpendicular magnetic fields $B \leq 27$ mT, which are more than one order of magnitude smaller than for reference samples without antidots. The presented results may also be relevant for other superconducting thin film microelectronic devices, e.g., kinetic inductance detectors, filters, or quantum interference devices, when operated in external magnetic fields.

The fabrication process starts with dc magnetron sputtering of a ~150 nm thick Nb layer onto a 2″ r-cut sapphire wafer. These Nb films have a critical temperature $T_c \approx 9$ K and a residual resistance ratio of $R_{300K}/R_{10K} \approx 6$. After cutting the wafer into single chips, lithography was performed. We covered the chips with photoresist and transferred a monolayer of water suspended polystyrene microspheres onto them in a Langmuir-Blodgett deposition process. The microspheres have an average diameter of 770 nm (± 25 nm) and act as microlenses for UV-light, leading to a focused energy density directly below each sphere and, thus, after development to a quasihexagonally arranged hole pattern. With reactive ion etching (SF_6), the hole array was transferred into the Nb film.

Electric transport measurements on test structures with antidots, which were performed close to T_c, showed pronounced matching effects. From these measurements, we estimate the antidot density to $n_p \approx 1.65\,\mu\text{m}^{-2}$. A detailed discussion of these results together with a deeper analysis of the properties of the self organized quasi-hexagonal antidot lattice will be given elsewhere.

Into the perforated as well as plain Nb films, we patterned half wavelength transmission line resonators with a

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resonance frequency $f_{res} \approx 6.25$ GHz using standard optical lithography and reactive ion etching (SF_6). The resonators are capacitively and symmetrically coupled to feed lines via gaps at both ends. For this study, we fabricated two types of resonators with the same general layout and length, but with different widths of the center conductor ($S_1 = 20\,\mu$m, $S_2 = 60\,\mu$m), of the gap between center conductor and ground planes ($W_1 = 8\,\mu$m, $W_2 = 25\,\mu$m) and of the coupling gaps ($G_1 = 10\,\mu$m, $G_2 = 30\,\mu$m). Both designs result in a characteristic impedance of the transmission line $Z_0 \approx 50\,\Omega$. Figure 1 shows (a) the resonator layout (parameter set 2) and (b) a partial AFM image of a resonator with antidots (parameter set 1).

For the characterization of the resonators, we mounted each of them into brass boxes and contacted them with silver paste to sub-miniature A (SMA) stripline connectors (signal line/center conductor) and the box (ground planes). We determined the frequency dependent transmitted power P_{trans} with a spectrum analyzer for different values of perpendicularly applied magnetic field $|B| \leq 27$ mT at a temperature $T = 4.2$ K. No attenuators or amplifiers were used in the measurements, but due to cable and connector losses, we estimate the effective power at the resonator input to be at least 10 dB lower than at the generator output, to which all values of the variable applied power P_{app} refer. We fit each resonance curve $P_{trans}(f)$ with a Lorentzian, from which we extract the resonance frequency $f_{res}(B)$ and the width of the resonance at half maximum $\Delta f(B)$. This allows us to calculate the quality factor $Q(B) = f_{res}(B)/\Delta f(B)$ and, alternatively, the energy loss $1/Q(B)$. In general, the overall losses $1/Q(B)$ are the result of different (field independent and field dependent) loss mechanisms such as dielectric, resistive, or radiative losses. The quantity $1/Q_v(B) = 1/Q(B) - 1/Q(0)$ allows us to extract the field dependent contribution, which we associate with Abrikosov vortices, cf. also Refs. 16 and 21. Similarly, we calculate the relative frequency shift due to the magnetic field $\delta f(B)/f_0 = (f_0 - f_{res}(B))/f_0$ with $f_0 = f_{res}(B = 0)$.

In one measurement series, we performed field cooling (fc) experiments, i.e., we apply each magnetic field value at a temperature above T_c and, subsequently, cool down the resonator below T_c. With this procedure, the vortices are expected to be nearly homogeneously distributed within the film.

Figure 2 compares (a) $1/Q_v(B)$ and (b) $\delta f(B)/f_0$ for the fundamental mode $n = 1$ of two resonators with identical parameters (set 2): one with and one without antidots. The resonator with antidots shows more than one order of magnitude lower losses (maximum factor ~40) and an about one order of magnitude reduced frequency shift (maximum factor ~11) than its plain counterpart over the whole investigated field range. We attribute this strong reduction of the vortex response to an effective trapping and pinning of vortices by the antidots. Note, that the first data point in Fig. 2 corresponds to $B = 0.45$ mT, i.e., to about 10 times Earth magnetic field, which is already well above the field range investigated in Ref. 18. In what we refer to as zero field here, that is, in Earth magnetic field and possible remanent fields in our setup, the resonators showed a comparable absolute quality factor of several thousands.

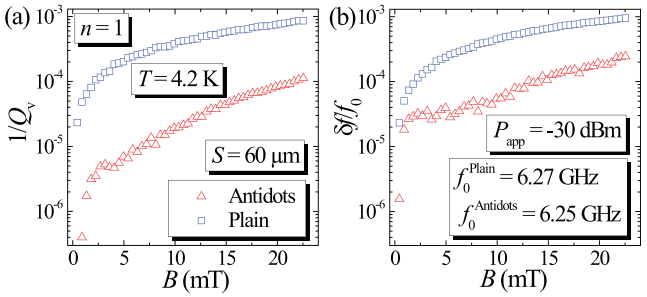


FIG. 2. (Color online) Comparison of (a) the vortex associated energy losses $1/Q_v(B)$ and (b) the relative frequency shift $\delta f(B)/f_0$ in fc measurements for the fundamental mode $n = 1$ of a perforated resonator (triangles) and a plain resonator (squares) with identical design parameters (set 2).

We also performed measurements for higher resonator modes, i.e., $n = 2,3$, and for resonators with parameter set 1 which all showed similar results. At higher frequencies, however, we were only able to reliably determine the losses for a somewhat smaller field range ($\lesssim 15$ mT) than for $n = 1$ due to a combination of suppressed cavity resonances and the appearance of parasitic ones.

In another set of experiments, resonators were cooled down to $T = 4.2$ K in $B = 0$ before applying a magnetic field. Here, the vortices are expected to penetrate the superconductor with a Bean-like flux profile,[27] which leads to an *enhanced* flux density close to the edges of the Nb structures. Figure 3 shows the vortex associated losses $1/Q_v(B)$ of (a) two resonators with parameter set 1 and (b) two resonators with parameter set 2. Within each parameter set, we compare a resonator with antidots to a plain one.

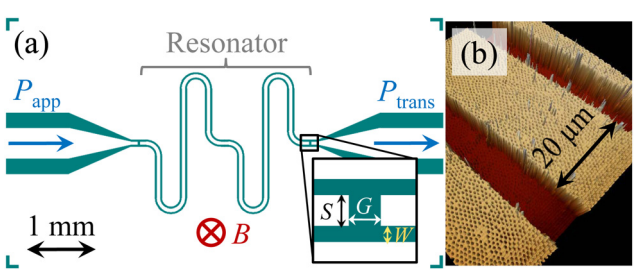


FIG. 1. (Color online) (a) Layout of a 7×4 mm^2 chip with a capacitively coupled 6.25 GHz transmission line resonator (parameter set 2). (b) AFM image of a part of a resonator perforated with antidots (parameter set 1).

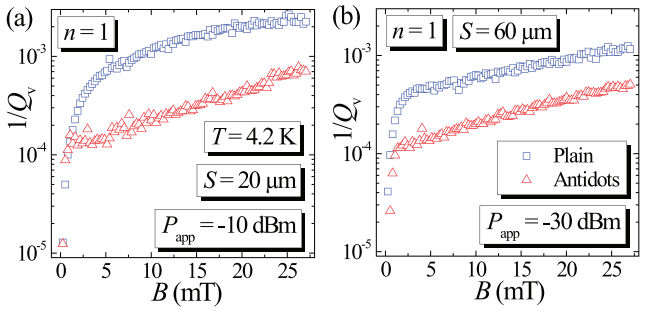


FIG. 3. (Color online) Vortex associated energy losses $1/Q_v(B)$ for zfc measurements of two resonators with (a) parameter set 1 and (b) parameter set 2. One resonator of each parameter set was patterned with antidots (red triangles), and one was not (blue squares).

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We find higher losses $1/Q_v(B)$ than in the corresponding fc measurements. This result, which *prima facie* is somehow counterintuitive, since the number of vortices within the superconductor is smaller for the zfc than for the fc case, has already been reported before.[28,29] It can be explained by the inhomogeneous flux distribution for the zfc situation which has its maximum near the Nb film edges. Since the microwave current density is also higher at the edges, this leads to increased losses $1/Q_v(B)$ when compared to the field-cooled case.

In the zfc case, for both parameter sets, the resonator with antidots shows significantly reduced losses $1/Q_v(B)$ over the whole field range (up to a factor of about 6). Yet, the absolute values of $1/Q_v(B)$ are higher for resonators with parameter set 1 than those of parameter set 2. There are two possible mechanisms, which may explain this observed behavior. As parameter set 1 corresponds to smaller gaps between the center conductor and the ground planes, the flux focusing and, hence, the effective flux density seen by the resonator are higher. The other mechanism reveals itself by occasionally observable sudden jumps in the transmitted power corresponding to a small shift of the resonance curve which are an indication for the appearance of flux avalanches[29] (also represented in the noise of $1/Q_v(B)$). Those flux avalanches reduce the flux density at the conductor edges by transporting flux towards the center of the conductor, where the microwave current density is considerably smaller. Obviously, the flux can be transported farther away from the edges for the case of a wider center conductor. Therefore, flux avalanches in resonators with parameter set 2 may result in a stronger flux reduction at the edges of the center conductor than in resonators with parameter set 1.

In conclusion, we have used microsphere photolithography for the fabrication of large area periodic micron-sized antidot arrays in superconducting thin film microwave devices to significantly improve the device performance in magnetic fields. We have demonstrated, that energy losses in superconducting microwave resonators due to the presence of Abrikosov vortices can be reduced by more than one order of magnitude with quasihexagonal antidot arrays. We have shown this result to be valid for magnetic fields up to at least $B \approx 27\,\mathrm{mT}$ for fc procedures as well as for zfc measurements.

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